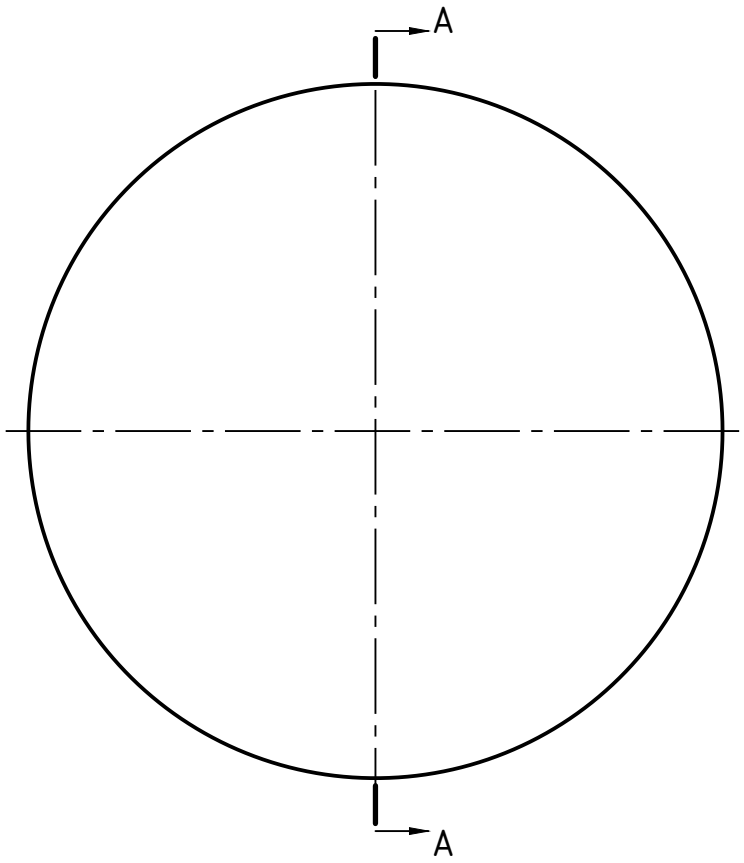
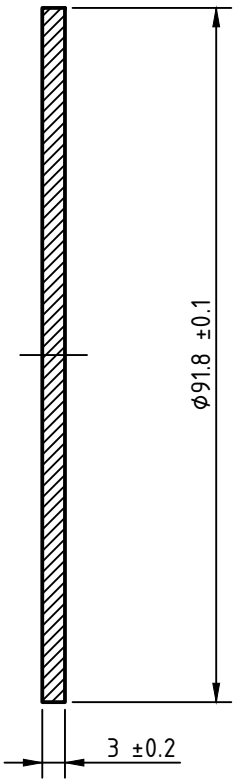
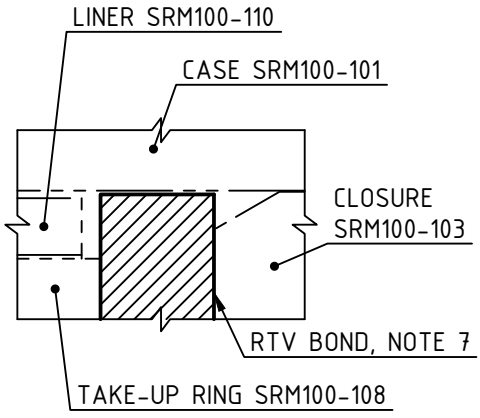




SECTION A–A

NOTES

1. Material: EPDM, peroxide cured, no fabric insert, to ASTM D2000 M2DA610; certificate required.
2. Stock: sheet 3 ±0.2, smooth faces, no release agent or bloom. The thickness does not stack: the take-up gap is measured at assembly (SRM100-000 sheet 3).
3. Cut to fwd_insulator.dxf (1:1, mm, one OD circle on layer CUT): full disc, no hole; pure water jet or CNC oscillating knife. No laser. Edges square, no tears or nicks.
4. ISO 10579-NR: free state, lying flat, at 23 °C.
5. Inspect 100 %: OD at 2 × 90° by template or caliper, no squeeze; thickness at 4 points, dial gauge, flat foot, light load. No cuts or voids.
6. Fit in the forward seal zone of case SRM100-101, honed to A +0.030/0, A ≤ 92.070: bore ø92.000 … 92.10, diametral clearance 0.10 … 0.40.
7. Bond after the hydrostatic proof to the aft face of closure SRM100-103: abrade both faces P180, IPA wipe, silicone primer SS4004P or equal, RTV106 or equal over the whole face, bondline 0.1 – 0.2; OD centred on the land within 0.1. Cure 24 h, 23 °C.



INSTALLED (5:1) – REFERENCE

8. At assembly grease the OD edge: Parker Super O-Lube or DC-55, thin film.
9. Replace after every firing. Bag label: SRM100-107, batch, cure date; do not mark the part.
10. USED ON: SRM100-000 motor, item 7 (1 off); not fitted in SRM100-001.

MATERIAL	MASS	GENERAL TOLERANCES	SURFACE TEXTURE	EDGES
EPDM, 60 ±5 Shore A ASTM D2000 M2DA610, sheet 3	22.8 g	ALL TOLERANCED	–	SQUARE, AS CUT

DOCUMENT TYPE	DOCUMENT STATUS	SCALE	PROJECTION	UNITS
DETAIL DRAWING	NOT CHECKED – FOR REVIEW	1:1		mm

LEGAL OWNER	TITLE, SUPPLEMENTARY TITLE
HELIOS SRM-100 SOLID ROCKET MOTOR	FORWARD INSULATOR EPDM disc on the forward closure face – cut from 3 sheet

CREATED BY	APPROVED BY	DATE OF ISSUE	IDENTIFICATION NUMBER	REV	SHEET
text-to-cad / Claude	–	2026-09-23	SRM100-107	A	1 / 1